

Why Data Testing



Why do we need ETL Testing?

Data is one of the most important asset of any enterprise.

Maintaining data quality is to ensure the success in this information where there are terabytes and petabytes of data created every second across the world over the internet and enterprise level.

A producer wants to know....



How **DATA QUALITY** impacts **BUSINESS PROCESS**

53%
Companies that have **SUFFERED LOSSES, PROBLEMS OR COSTS** due to poor data quality



57%
Increase in **CUSTOMER SATISFACTION** if a company has high-quality, business-driven data

The main reason cited for Data Migration projects running late and/or over budget:

POOR DATA QUALITY



According to the report there businesses who are losing close to 8 million every year due to bad data.

What is bad data?

Bad data is an inaccurate set of information, including missing data, wrong information, inappropriate data, non-conforming data, duplicate data and poor entries (misspells, typos, variations in spellings, format etc.).

How to deal with bad data?

By using etl process we will improve data quality.

Why do we need ETL process?

The agenda of ETL process is to achieve business Intelligence

Business Intelligence is the process of collecting raw data or business data and turning it into information that is useful and more meaningful. The raw data is the records of the daily transaction of an organization such as interactions with customers, administration of finance, and management of employee and so on.

These data's will be used for "Reporting, Analysis, Data mining, Data quality and Interpretation, Predictive Analysis".

Need of ETL testing specifically

ETL is commonly associated with any of the data warehousing projects. But in reality any form of data movement from a source to a target system can be considered as ETL. Large enterprises quite often have need to move application data from source to another or one application to another for data integration or data migration purposes.

What is etl testing?

ETL Testing is data centric testing process where we need to validate that data has been transformed and loaded into the target system as expected.

It is a testing method in which the data inside a data warehouse is tested for integrity, reliability, accuracy and consistency in order to comply with the company's data framework. The main purpose of data warehouse testing is to ensure that the integrated data inside the data warehouse is reliable enough for a company to make decisions on.

Challenges in etl testing /what are the challenges you have faced during your etl testing?

ETL testing is quite different from conventional testing. There are many challenges we faced while performing data warehouse testing.

1) Incompatible and duplicate data.

Incompatible refers to invalid data or under defined.

Generally in source system there are duplicate data, which needs to treat properly before loading it into the warehouse without data loss.

2) Loss of data during ETL process.

Due to transformation logic we see loss of data that could be a valid drop or invalid drop.

In case of invalid drop we need to check and verify drop reason,
These details you can get from your log table.

3) Testers have no privileges to execute ETL jobs by their own.

4) Volume and complexity of data is very huge.

5) Missing business flow information.

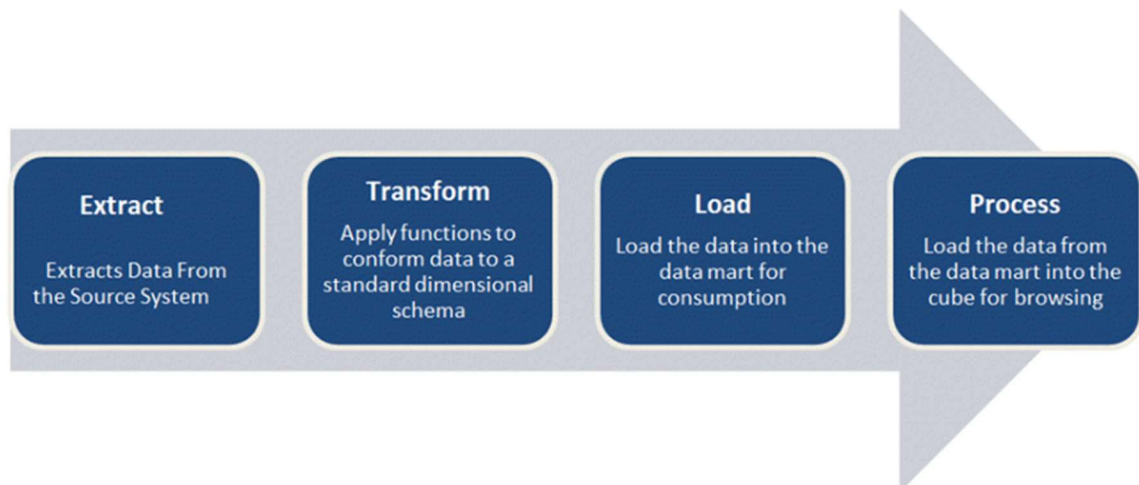
Heterogeneous data

Heterogeneous means different data types and formats.

Heterogeneous data are any data with high variability of data types and formats. They are possibly ambiguous and low quality due to missing values, high data redundancy, and untruthfulness. It is difficult to integrate heterogeneous data to

meet the business information demands. For example, heterogeneous data are often generated from Internet of Things (IoT).

ETL Process



The data in a Data Warehouse system is loaded with an ETL (Extract, Transform, and Load) tool. As the name suggests, it performs the following three operations –

Extracting the Data

It involves extracting the data from different heterogeneous data sources.

Data extraction from a transactional system varies as per the requirement and the ETL tool in use.

It is normally done by running scheduled jobs in off-business hours like running jobs at night or over the weekend.

Transforming the Data

It involves transforming the data into a suitable format that can be easily loaded into a DW system. Data transformation involves applying calculations, joins, and defining primary and foreign keys on the data.

For example, if you want % of total revenue which is not in the database, you will apply % formula in transformation and load the data. Similarly, if you have the first name and the last name of users in different columns,

Then you can apply a concatenate operation before loading the data. Some data doesn't require any transformation; such data is known as direct move or pass through data.

Data transformation also involves data correction and cleansing of data, removing incorrect data, incomplete data formation, and fixing data errors. It also includes data integrity and formatting incompatible data before loading it into a DW system.

Data integrity-data accuracy and consistency over its entire life-cycle

Loading the Data into a DW System

It involves loading the data into a DW system for analytical reporting and information. The target system can be a simple delimited flat file or a data warehouse.

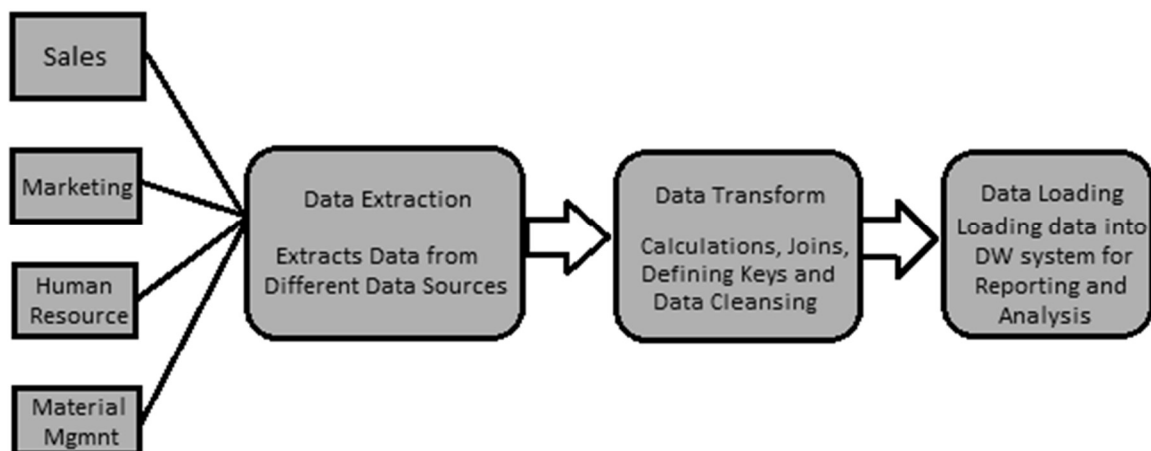
Later, we can use various Business Intelligence (BI) tools to generate meaningful reports, dashboards, and visualizations using this data.

Let us take an example to understand it better.

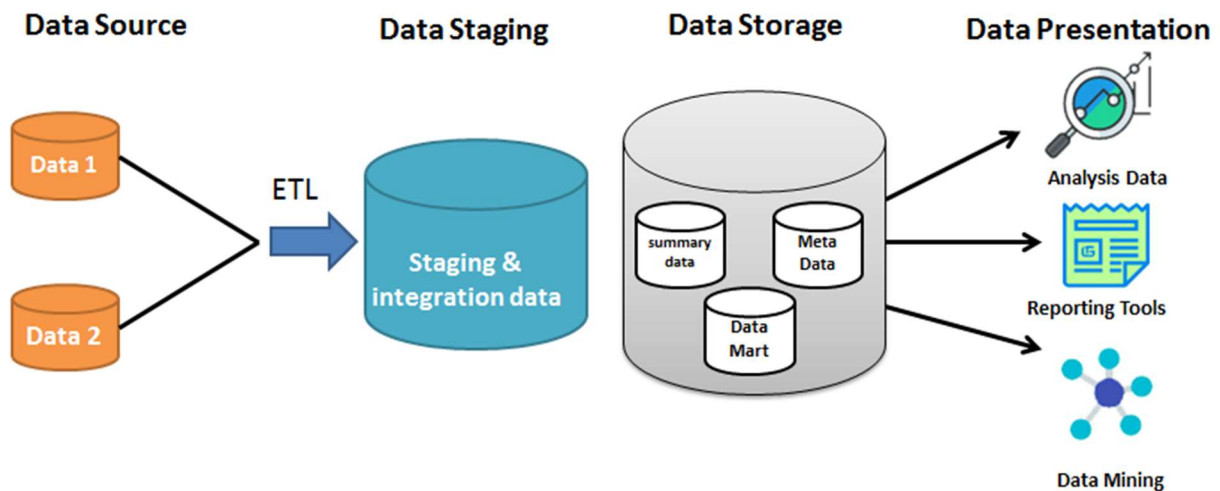
In case of a manufacturing company having multiple departments such as sales, HR, Material Management, EWM, etc. All these departments have separate databases which they use to maintain information with respect to their work and each database has a different technology, landscape, table names, columns, etc.

Now, if the company wants to analyze historical data and generate reports, all the data from these data sources should be extracted and loaded into a Data Warehouse to save it for analytical work.

An ETL tool extracts the data from all these heterogeneous data sources, transforms the data (like applying calculations, joining fields, keys, removing incorrect data fields, etc.), and loads it into a Data Warehouse.



ETL Layers /ETL Architecture



1. Data Source Layer

- The Data Source Layer is the layer where the data from the source is encountered and subsequently sent to the other layers for desired operations.
- The data can be of any type.
- The Source Data can be a database, a Spreadsheet or any other kind of text file.
- The Source Data can be of any format. However, we cannot expect to get data with the same format considering the sources are vastly different.
- In Real Life, Some examples of Source Data can be
 - Log Files of each specific application or job or entry of employers in a company.
 - Survey Data, Stock Exchange Data, etc.
 - Web Browser Data and many more.

2. Data Staging Layer

- The following steps take place in Data Staging Layer.

Step #1: Data Extraction

The Data received by the Source Layer is feed into the Staging Layer, where the first process that takes place with the acquired data is extraction.

Step #2: Landing Database

- 1) The extracted data is temporarily stored in a landing database.
- 2) It retrieves the data once the data is extracted.

Step #3: Staging Area

- 1) The Data in Landing Database is taken, and several quality checks and staging operations are performed in the staging area.
- 2) The Structure and Schema are also identified, and adjustments are made to data that are unordered, thus trying to bring about a commonality among the data that has been acquired.
- 3) Having a place or set up for the data just before transformation and changes is an added advantage that makes the Staging process very important.
- 4) It makes data processing easier.

Step #4: ETL

- 1) It is an Extraction, Transformation, and Load.

- 2) ETL Tools are used for the integration and processing of data where logic is applied to rather raw but somewhat ordered data.
- 3) This data is extracted as per the analytical nature that is required and transformed to data that is deemed fit to be stored in the Data Warehouse.
- 4) After Transformation, the data or rather information is finally loaded into the data warehouse.
- 5) Some examples of ETL tools are Informatica, SSIS, etc.

3. Data Storage Layer

- 1) The processed data is stored in the Data Warehouse.
- 2) This Data is cleansed, transformed, and prepared with a definite structure and thus provides opportunities for employers to use data as required by the Business.
- 3) Depending upon the approach of the Architecture, the data will be stored in Data Warehouse as well as Data Marts. Data Marts will be discussed in the later stages.
- 4) Some also include an Operational Data Store.

4. Data Presentation Layer

- 1) This Layer where the users get to interact with the data stored in the data warehouse.
- 2) Queries and several tools will be employed to get different types of information based on the data.
- 3) The information reaches the user through the graphical representation of data.
- 4) Reporting Tools are used to get Business Data, and Business logic is also applied to gather several kinds of information.
- 5) Meta Data Information and System operations and performance are also maintained and viewed in this layer.

Data Mart

An enterprise data warehouse can be divided into subsets, also called data marts, which are focused on a particular business unit or department. Data marts allow selected groups of users to easily access specific data without having to search through an entire data warehouse.

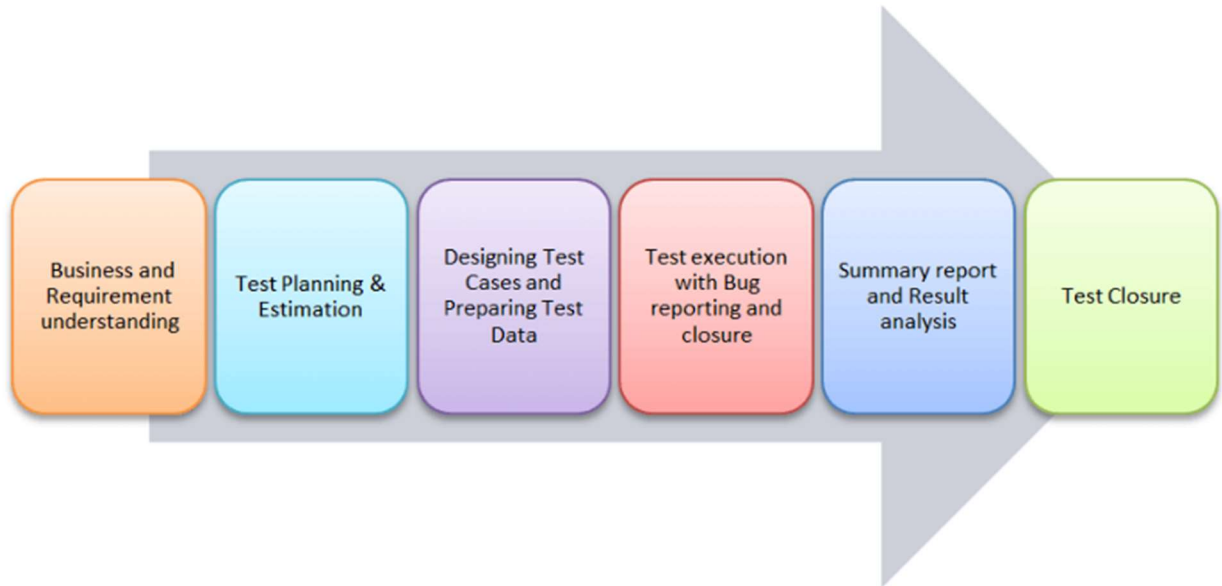
Why do we need staging layer?

- 1) The data in landing database is taken and several quality checks and data format checks are performed in the staging area.
- 2) The structure and schemas are also identified and adjustments are made to data that are unordered, thus trying to bring about a commonality among the data that has been acquired.
- 3) Having a place or set up for the data just before transformation and change is an added advantage that makes the staging process very important.
- 4) Temporary tables are also created in this layer to add additional columns on the fly.
- 5) It makes data processing easier.

ETL Tools

- 1) Informatica Power Center
- 2) IBI Data Migrator
- 3) Congas Data Manager
- 4) SQL Server Integration Services (SSIS)
- 5) SAP Data Services

ETL Testing Process



1) Requirement gathering

We have one document called Mapping sheet which contains information for all columns in a table with their transformation logic.

2) Define test scenarios

-Test scenarios will be covering both positive and negative scenarios

3) Test case preparation

-Preparing test case as per test scenarios.

4) Prepare test data as per mapping sheet.

Few of times we do not have any transformation from source to target, in that case it will direct dump.

5) Extract data and run test case.

- 6) Transform data and run tests.
- 7) Load data into the target databases and run test case.
- 8) Run end to end tests.
- 9) Generating a test report.

Scope of ETL Testing

1) Metadata Testing

- Table structure validation including data type testing.
- Data length check
- Constraint checks
- Naming convention issues

2) Data Quality check

-Checking data quality in terms of performs the following scenarios – Primary Key, Foreign Key, NOT NULL, NULL, and UNIQUE.

- Duplicate Data Check Testing

This testing involves checking for duplicate data in the target system. When there is a huge amount of data in the target system, it is possible that there is duplicate data in the production system that may result in incorrect data in Analytical Reports.

Duplicate values can be checked with SQL statement like –

```
SELECT COUNT(*),EMPLOYEE_ID FROM EMP  
GROUP BY EMPLOYEE_ID HAVING COUNT(*)>1
```

Note:-Here we are assuming EMPLOYEE_ID as PK column and we are expecting no duplicate data flows.

The result of this query should be zero.

Incase if we find duplicates we need to ask for composite primary key.

Duplicate data appears in the target system due to the following reasons –

- If no primary key is defined, then duplicate values may come.
- Due to incorrect mapping or environmental issues.
- Manual errors while transferring data from the source to the target system

- Else we can ask for composite primary key columns.

-Data quality testing involves performing number check, date check, null check, precision check, etc. A tester performs Syntax Test to report invalid characters, incorrect upper/lower case order, etc.

3) Data Completeness Testing

-Row count validation

It involves checking the count of data in the source and the target systems. It doesn't involve checking the values of data in the target system. It also doesn't involve if the data is in ascending or descending order after mapping of data.

-Column level data validation

It checks the data values in the source system and the corresponding values in the target system after transformation. This type of testing is time-consuming and it requires broader way

Where we need to cover all columns which are correctly mapped with our source system

Columns.

-Data Consistency Issues

The data type and length for a particular attribute may vary in files or tables though the semantic definition is the same.

Example: Account number may be defined as: Number (9) in one field or table and Varchar2 (11) in another table

-Check Data should not be truncated in the column of target table.

-Check boundary value analysis (ex: only ≥ 2008 year data has to load into the target)

4) Data Correctness Issues

-Data that is misspelled or inaccurately recorded.

-Null, non-unique, or out of range data may be stored when the integrity constraints are disabled.

Example: The primary key constraint is disabled during an import function. Data is entered into the existing data with null, unique identifiers.

5) Null Validation

-Verify the null values where "Not Null" specified for specified column.

5) History Load Testing

-History load testing is done to ensure that one time or past dated records has loaded as expected.

-We need to ensure there is no invalid record drop while data load process.

-History load testing will done on some specific date range.

6) Incremental Testing

Incremental testing is performed to verify if Insert and Update statements are executed as per the expected result. This testing is performed step-by-step with old and new data.

7) Regression Testing

When we make changes to data transformation and aggregation rules to add new functionality which also helps the tester to find new errors, it is called Regression Testing. The bugs in data that comes in regression testing are called Regression.

1) Retesting

When you run the tests after fixing the codes, it is called retesting.

2) System Integration Testing

System integration testing involves testing the components of a system individually and later integrating the modules. There are three ways a system integration can be done: top-down, bottom-up, and hybrid.

8) Complete Data Validation (using minus and intersect)

1. To validate the complete data set in source and target table minus query is best solution.
2. We need to source minus target and target minus source.
3. If minus query returns any value those should be considered as mismatching rows.
4. We need to match rows among source and target using Intersect statement.
5. The count returned by intersect should match with individual counts of source and target tables.
6. If minus query returns 0 rows and count intersect is less than source count or target table count then we can considered as duplicate rows are exists.

Roles of ETL Testing

Since ETL testing is so important, ETL testers are in great demand. ETL testers validate data sources, extract data, apply transformation logic, and load data into target tables. The following are key responsibilities of an ETL tester:

1. In-depth knowledge of ETL tools and processes.

2. Performs thorough testing of the ETL software.
3. Check the data warehouse test component.
4. Perform the backend data-driven test.
5. Design and execute test cases, test plans, test harnesses, etc.
6. Identifies problems and suggests the best solutions.
7. Review and approve the requirements and design specifications.
8. Writing SQL queries for testing scenarios.
9. Various types of tests should be carried out, including primary keys, defaults, and checks of other ETL-related functionality.
10. Conducts regular quality checks

ETL Test Scenarios

Test Scenarios	Test-Cases
Structure Validation	It involves validating the source and the target table structure as per the mapping document.
	Data type should be validated in the source and the target systems.
	The length of data types in the source and the target system should be same.
	Data field types and their format should be same in the source and the target system.
	Validating the column names in the target system.
Validating Mapping document	It involves validating the mapping document to ensure all the information has been provided. The mapping document should have change log, maintain data types, length, transformation rules, etc.
Validate Constraints	It involves validating the constraints and ensuring that they are applied on the expected tables.
Data Consistency check	It involves checking the misuse of integrity constraints like Foreign Key.
	The length and data type of an attribute may vary in different tables, though their definition remains same at the semantic layer.
Data Completeness Validation	It involves checking if all the data is loaded to the target system from the source system.
	Counting the number of records in the source and the target systems.
	Boundary value analysis.
	Validating the unique values of primary keys.
	It involves validating the values of data in the target system.
	Misspelled or inaccurate data is found in table.

Data Correctness Validation	Null, Not Unique data is stored when you disable integrity constraint at the time of import.
Data Transform validation	It involves creating a spreadsheet of scenarios for input values and expected results and then validating with end-users.
	Validating parent-child relationship in the data by creating scenarios.
	Using data profiling to compare the range of values in each field.
	Validating if the data types in the warehouse are same as mentioned in the data model.
	It involves performing number check, date check, precision check, data check, Null check, etc. Example – Date format should be same for all the values.
Null Validation	It involves checking the Null values where Not Null is mentioned for that field.
Duplicate Validation	It involves validating duplicate values in the target system when data is coming from multiple columns from the source system.
	Validating primary keys and other columns if there is any duplicate values as per the business requirement.
Date Validation check	Validating date field for various actions performed in ETL process.
	Common test-cases to perform Date validation –
	From Date should not greater than To Date
	Format of date values should be proper.
	Date values should not have any junk values or null values
Full Data Validation Minus Query	It involves validating full data set in the source and the target tables by using minus query.
	You need to perform both source minus target and target minus source.
	If the minus query returns a value, that should be considered as mismatching rows.
	You need to match the rows in source and target using the Intersect statement.
	The count returned by Intersect should match with the individual counts of source and target tables.
	If the minus query returns no rows and the count intersect is less than the source count or the target table count, then the table holds duplicate rows.
Other Test Scenarios	Other Test scenarios can be to verify that the extraction process did not extract duplicate data from the source system.
	The testing team will maintain a list of SQL statements that are run to validate that no duplicate data have been extracted from the source systems.
Data Cleaning	Unwanted data should be removed before loading the data to the staging area.

What is ELT?

ELT is the process by which raw data is extracted, loaded, and transformed into a data lake or warehouse. In contrast to ETL, ELT provides faster loading.

ELT vs. ETL

The differences between ELT and a traditional ETL process are more significant than just switching the L and the T. The biggest determinant is how, when and where the data transformations are performed.

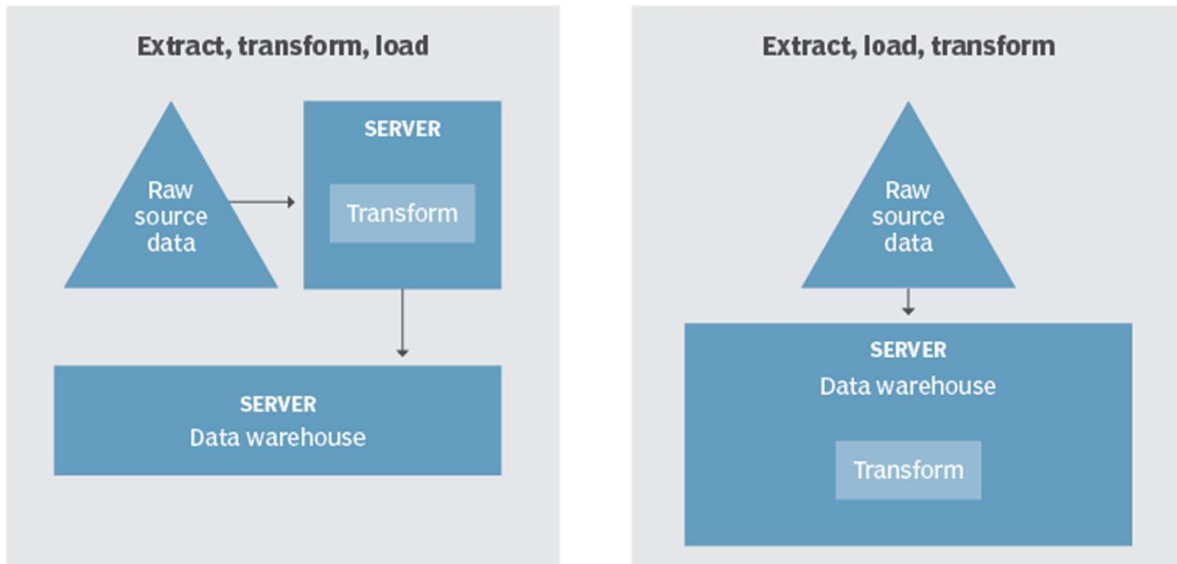
With ETL, the raw data is not available in the data warehouse because it is transformed before it is loaded. With ELT, the raw data is loaded into the data warehouse (or Data Lake) and transformations occur on the stored data.

Data Lake- A Data Lake is a centralized repository designed to store, process, and secure large amounts of structured, semi structured, and unstructured data. It can store data in its native format and process any variety of it, ignoring size limits.

Staging areas are used for both ELT and ETL, but with ETL the staging areas are built into the ETL tool being used. With ELT, the staging area is in a database used for the data warehouse.

ETL vs. ELT

ETL and ELT differ from each other when it comes to where data processing occurs, as illustrated below.



The following chart compares different facets of ETL or ELT:

	ETL	ELT
Source Data	Support storing structured data from input sources	Can be used for structured, unstructured, and semi-structured data types
Data Size	Best suited for smaller amounts of data	Can be used for large amounts of data
Storage Type	Can be used for on-premises or cloud storage	Optimized for cloud data warehouses
Latency	High, as transformations need to be completed before storing data	Low, as minimal processing is done before storing in the data warehouse
Flexibility	Low, as data sources and transformations need to be defined at the beginning of the process	High, as transformation need not be defined when integrating new sources
Scalability	Can be low, as the ETL tool should support scaling of operations	High, as ELT tools can be easily configured for changing data sources
Maintenance	May need continuous maintenance in case of changes in data sources or formats	Low maintenance required as usually ELT tools automate the process

Compliance with Security Protocols	Easy to Implement	May need to be supported by data warehouse/ELT tool
Storage Requirement	Low as only transformed data is stored	Can be high as raw data is stored

Benefits of ELT

One of the main attractions of ELT is the reduction in load times relative to the ETL model. Taking advantage of the processing capability built into a data warehousing infrastructure reduces the time that data spends in transit and is usually more cost-effective. ELT can be more efficient by utilizing the computer power of modern data storage systems.

When you use ELT, you move the entire data set as it exists in the source systems to the target. This means that you have the raw data at your disposal in the data warehouse, in contrast to the ETL approach where the raw data is transformed before it is loaded to the data warehouse. This flexibility can improve data analysis, enabling more analytics to be performed directly within the data warehouse without having to reach out to the source systems for the untransformed data.

Using the ELT can make sense when adopting a big data initiative for analytics. Big data often relies on a large amount of data, as well as wide variety of data that is more suitable for ELT.

Why ELT is required:-

- 1) In case of unstructured source data.
- 2) ELT process is mainly used for cloud based source system.
- 3) ELT reduces cost compared to ETL.
- 4) Supports storing unstructured data and performing fast and large-scale data transformations.

ELT tools and software

Although ELT can be performed using separate tools for extracting, loading and transforming the data, tools exist that integrate all ELT processes. When seeking an ELT tool, users should look for the ability to read data from multiple sources, specifically the sources that their organization uses and intends to use. Most tools support a wide variety of source and target data stores and database systems.

Users can look for tools that can perform both ETL and ELT, as it's likely to have the need for both data integration techniques.

Although there are many ELT/ETL tool providers, a few of the market leaders include:

- 1) IBM
- 2) Informatica
- 3) Microsoft
- 4) Oracle
- 5) SAS
- 6) Talend
- 7) Teradata

ELT Process

